Holistic AI

UCL CENTRE FOR DIGITAL INNOVATION powered by aws

Can a web agent learn when a screenshot is worth the cost?

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Hindsight reveals a real routing opportunity. Against always-cheapest, none of eight learned routers — and only one of eight hindsight oracles — win on both success and cost. These agents solve just 2–36% of tasks, starving the router of the supervision it would need.

IN HINDSIGHT

+16.35 pp

MORE TASKS SOLVED PER 100
vs the best single fixed mode

So there is something genuinely worth routing to.

WHEN ACTUALLY LEARNED

0 of 8

LEARNED ROUTERS
beat always-cheapest on both success and cost

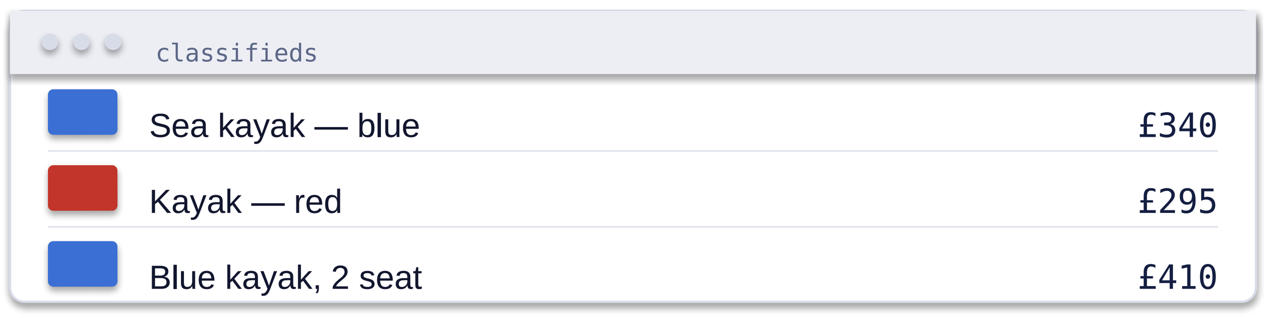
Not one beat always-cheapest on both counts.

WHAT WE ACTUALLY DID

THE AGENT IS GIVEN

"Find me the cheapest blue kayak on this site."

A REAL TASK — VISUALWEBARENA, CLASSIFIEDS #0



ILLUSTRATIVE — NOT A BENCHMARK SCREENSHOT

"blue" can be visual. **"cheapest"** is textual.

Should the agent pay to look?

RUN THE SAME TASK SIX WAYS

NO IMAGE AT ALL

DOM · P-text · P-prompt · P-SoM

TEXT + SCREENSHOT

SoM

SCREENSHOT ONLY

Vision

MEASURE SUCCESS × COST, THEN COMPARE

FIXED MODE
the same choice every task

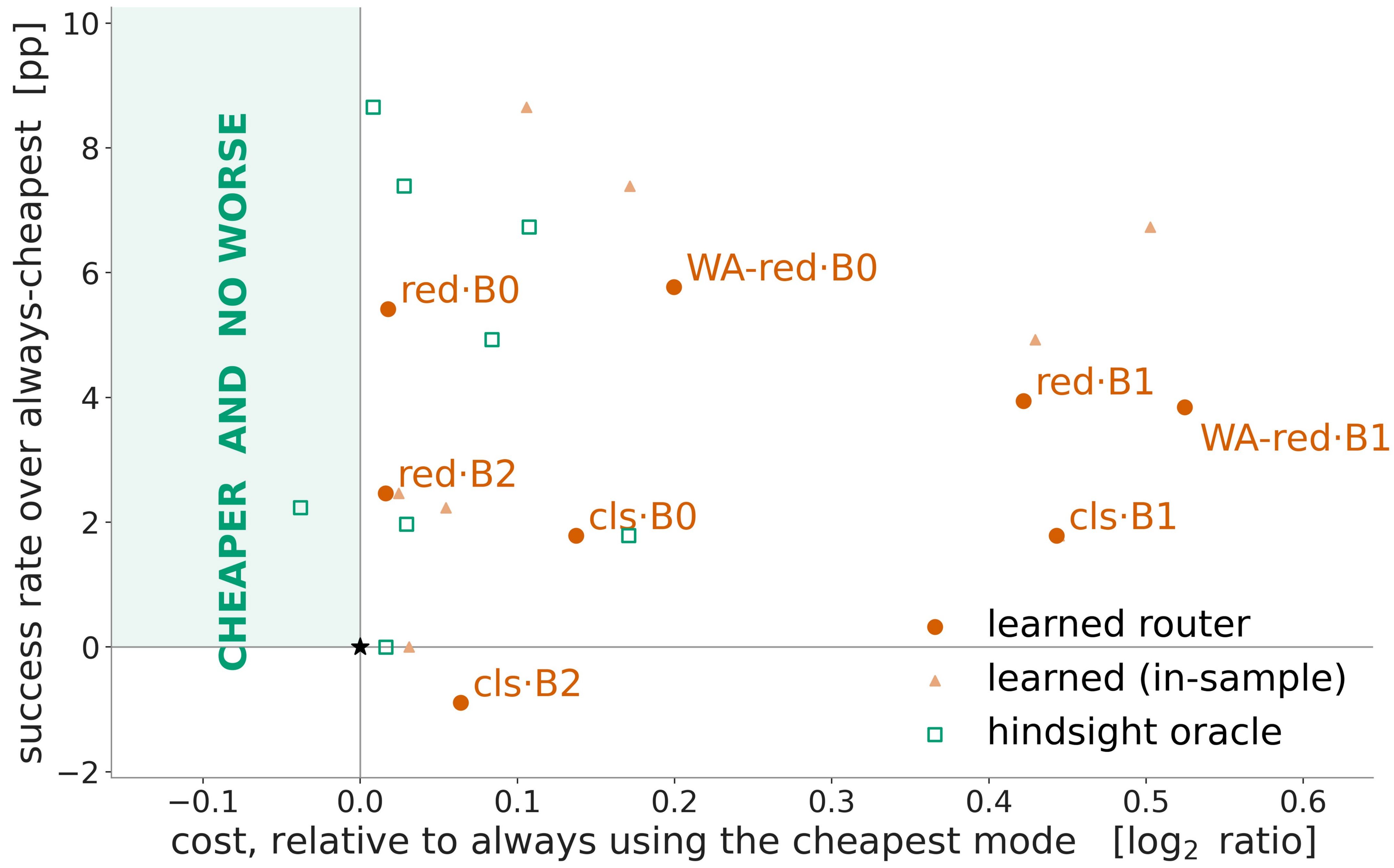
HINDSIGHT ORACLE
the best choice, known afterwards

LEARNED ROUTER
a choice made before the task runs

6 modes · 8 benchmark–model settings · 8,934 episodes

NOTHING WE TRAINED WON

Every policy, in every pair, measured against one simple fixed baseline: **always use the cheapest mode**. A win means landing in the shaded region — cheaper *and* no worse.



0 of 8 learned routers land in the win region.

1 of 8 hindsight oracles clear the same bar — so learning failure alone cannot explain this.

Always-cheapest is the cheapest fixed policy **on average**, not a per-episode floor — which is why a few points sit left of it. Nested cross-validation; 10,000 bundle permutations.

THE ROUTING PARADOX

Weak agents need routing most —
but produce the least supervision to learn it.

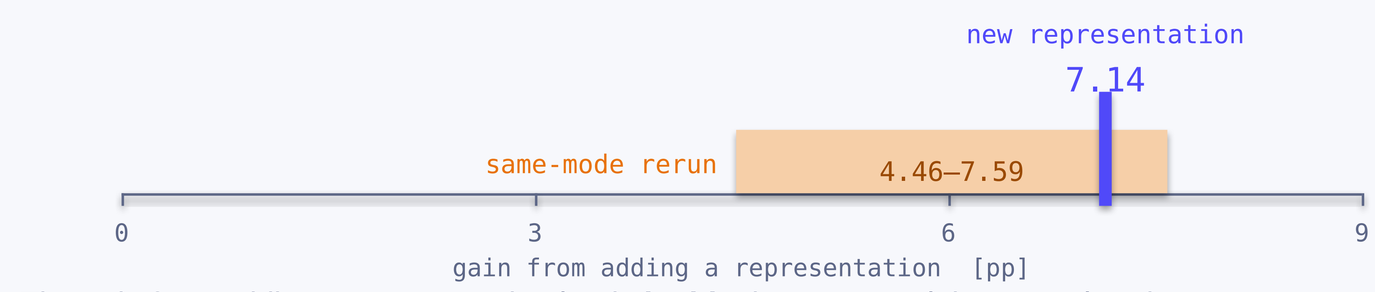
2–36%
OF TASKS THESE AGENTS SOLVE

15–97
USABLE ROUTING LABELS, TYPICALLY

A routing label is born only when the agent solves something.

CALIBRATION · WHAT SURVIVES

Rerunning **one mode on the same tasks** flips **12–14%** of outcomes. So we measured the ceiling against that ruler:



The gain from adding a representation lands **inside** the range a plain rerun already covers — not distinguishable here. B0 · classifieds, n=224, three replicated modes.

9.5–30.6% cheaper at identical success, in 8 of 8 pairs — send the tasks *nobody* solves to the cheapest mode.

Against the best-success fixed mode, and still a hindsight bound; in most pairs plain always-cheapest saves more. What makes it worth naming is the label: *solvable or not* is far easier to supply than *which mode*.

Routing is not only a model-selection problem. Its learnability depends on the competence of the agent producing the labels.

Measured inside the 2–36% success regime we observed. This conclusion need not hold for stronger agents.

SO, IN THIS ORDER

1. Improve the agent
2. Generate reliable supervision
3. Then learn selective perception

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Paper, code & full results: github.com/Quarkgluonmixture/Cost-Aware-Routing-for-Web-Usage-Agents

Explore all 8 settings →

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